VICTORY 2027

The Circulatory System

LECTURE-8 BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. Double Circulation
2. Hepatic Portal System
3. Lymph, Spleen


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

Double Circulation

Blood passes through the heart twice during one complete circulation.

Pulmonary Circulation

LUNGS

Right Ventricle → P. Artery → Lungs

Left Atrium ← P. Vein

P. Artery



Systemic Circulation

When Oxygenated blood from heart is sent to Body tissues or Cells.



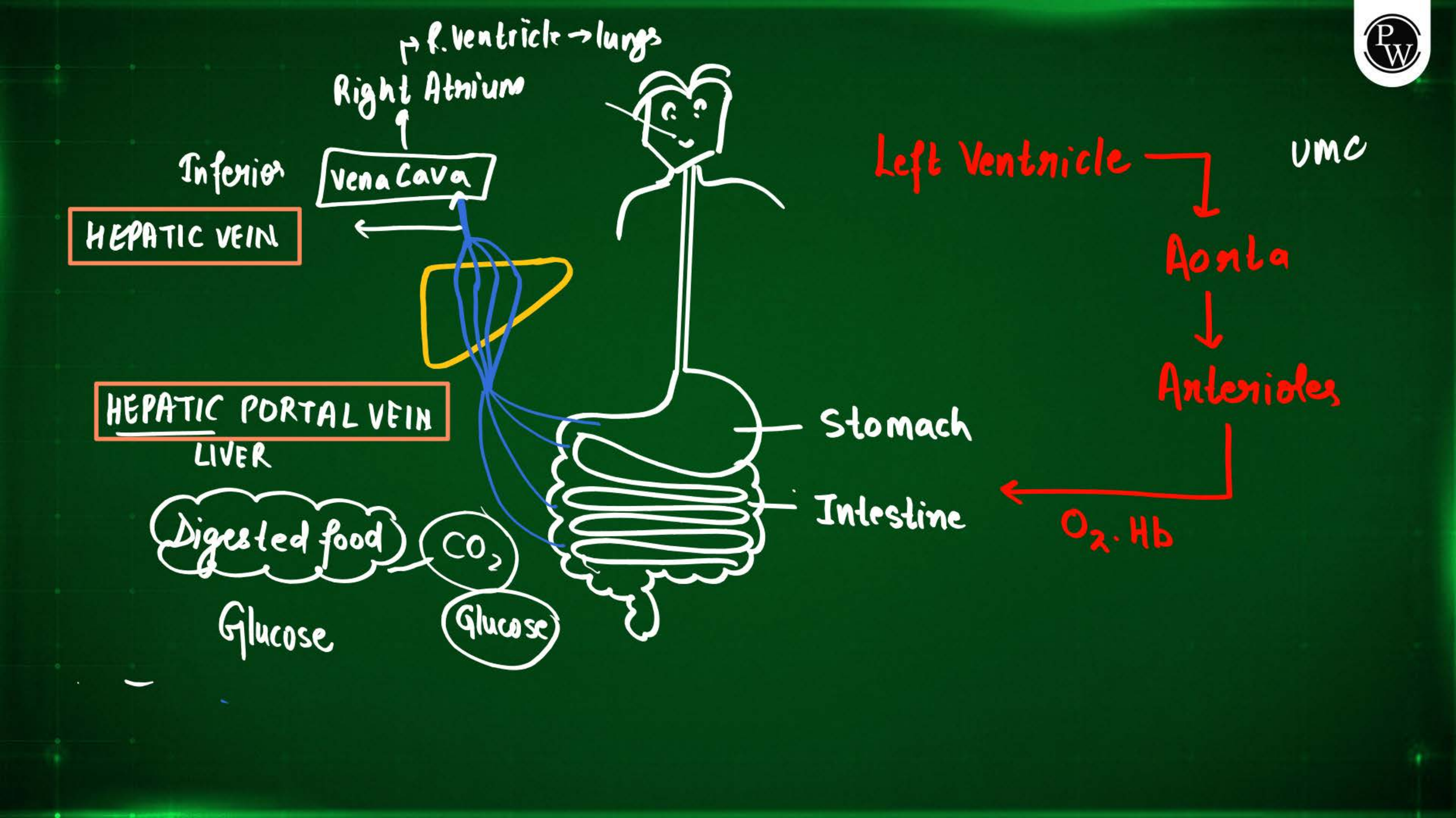
Left Ventricle → Aorta → Cells

Right Atrium ← Vena Cava

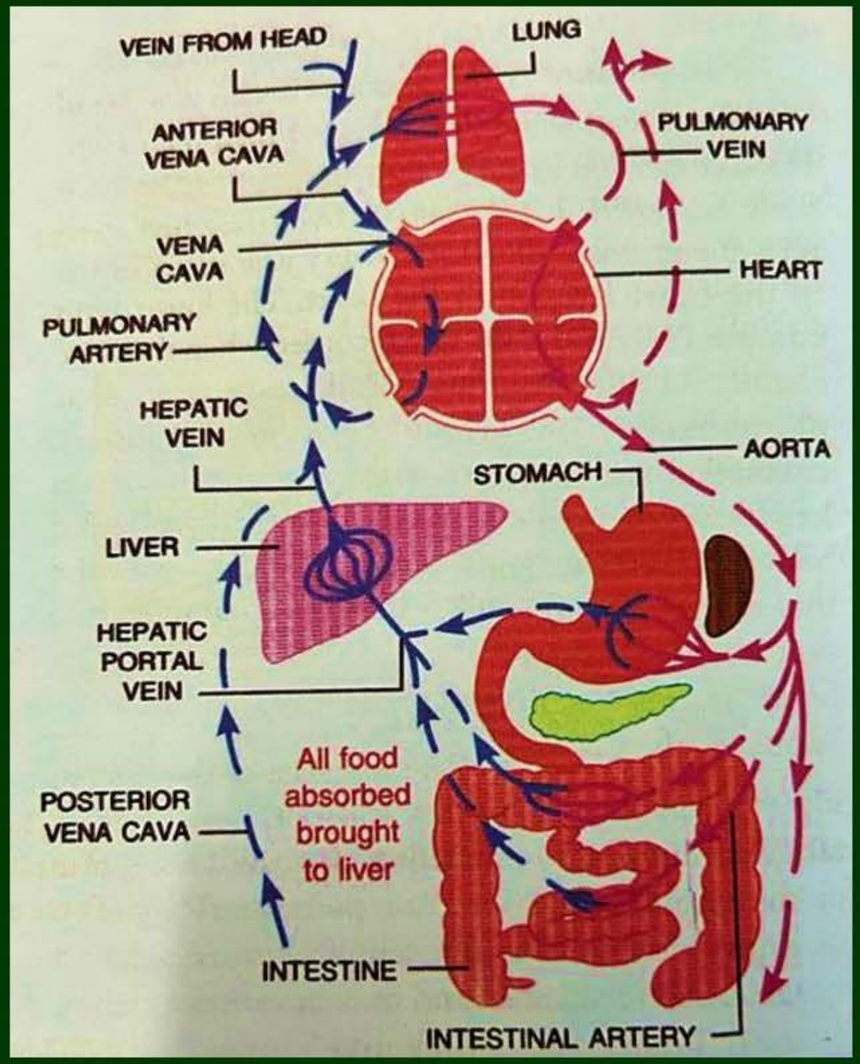


Hepatic Portal System

- The vein starts from stomach and intestine and do not directly convey the blood to posterior vena cava.
- Instead they first enter liver as combined hepatic portal vein.
- There it breaks up into capillaries and hepatic vein is formed by reunion which joins the posterior vena cava.
- Hence a portal vein is a vein which starts with capillaries and ends in capillaries.


R. Ventricle → lungs
Right Atrium
Inferior
Vena Cava
HEPATIC VEIN
HEPATIC PORTAL VEIN
LIVER
Digested food
CO_2
Glucose
Glucose
Stomach
Intestine
Left Ventricle
Aorta
Arterioles
O_2. Hb
UMC

Hepatic Portal System

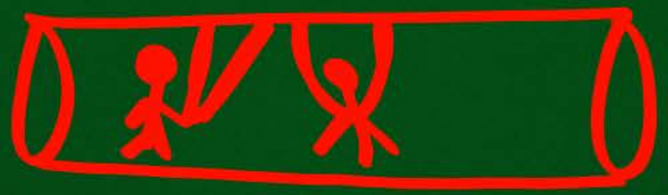


Blood pressure

- It is the pressure which the blood flowing through the arteries exerts on their walls.
- Upper limit (systolic pressure) – 120 mm Hg (contraction)
- Lower limit (diastolic pressure) – 80 mm Hg
- A rise in blood pressure above 140/90 is hypertension.
- Low in blood pressure than 140/90 is hypotension.
- Sphygmomanometer – instrument used to measure blood pressure.



120/80 mm Hg

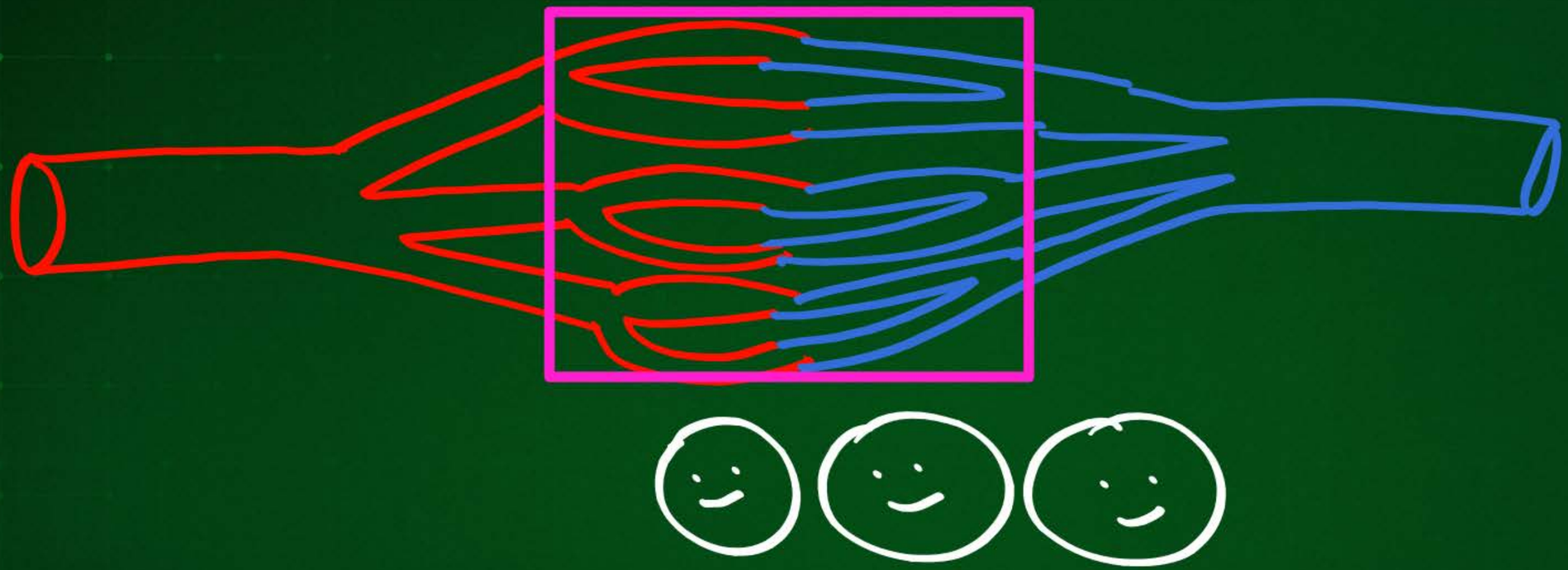


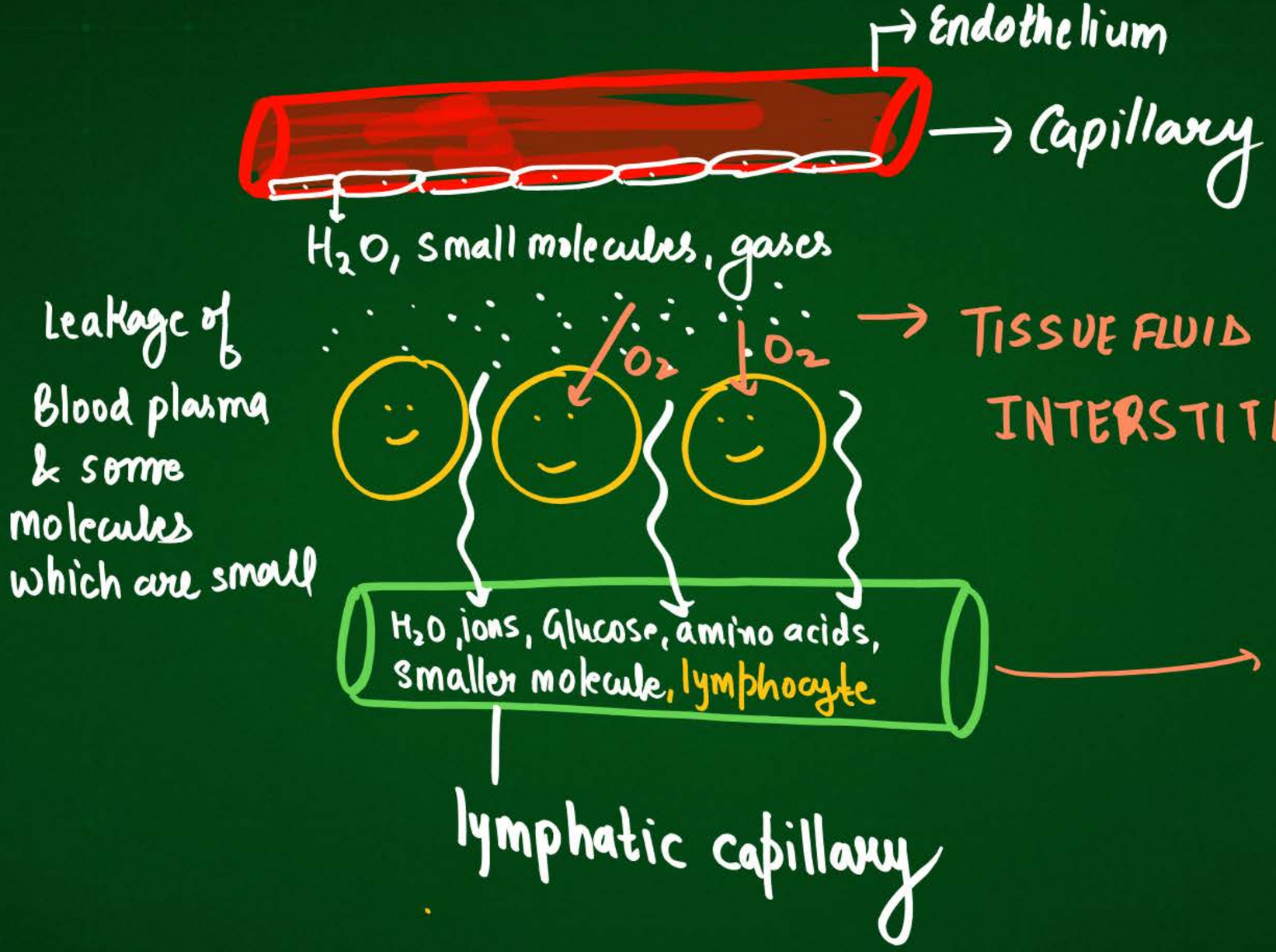




The pulse

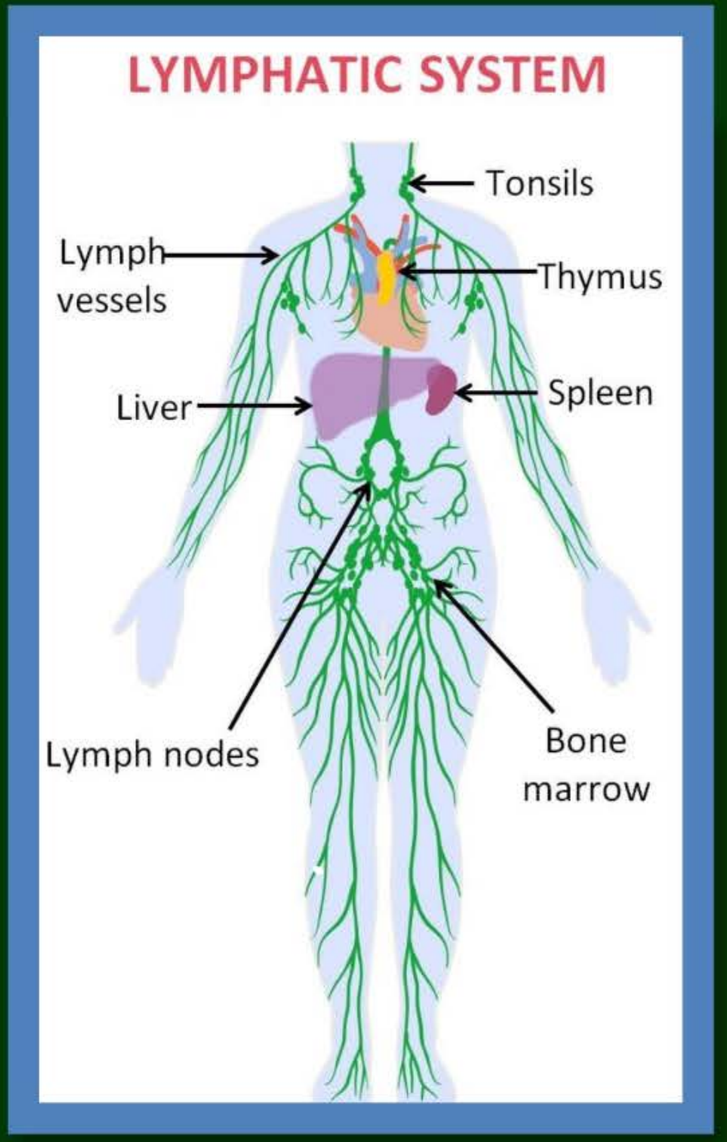
- It is the alternate expansion and elastic recoil of wall of artery during ventricular systole.
- Counting of pulse is indirectly the counting of heartbeat.




Endothelium
Capillary
H_2O, small molecules, gases
Leakage of Blood plasma & some molecules which are small
O_2
O_2
TISSUE FLUID
INTERSTITIAL FLUID
H_2O, ions, Glucose, amino acids, smaller molecule, lymphocyte
Mix into 'Vein' [Vena Cava].
lymphatic capillary

Lymph and lymphatic system

- Some tissue fluid gets reabsorbed into the blood vessels, and most of it enters another set of minute channels named lymph vessels where it is called lymph.
- Lymph vessels on the way drain lymph channels arise and ultimately pour lymph into the major anterior veins close to their entry into the right auricle and again circulation.





Composition of lymph

(a) Cellular part :

- Only leukocytes (mostly lymphocytes). No RBCs and blood platelets.



Composition of lymph

(b) Non-cellular part :

- Water – 94%.
- Solids (proteins, fats, carbohydrates, enzymes, antibodies, etc) – 6%.

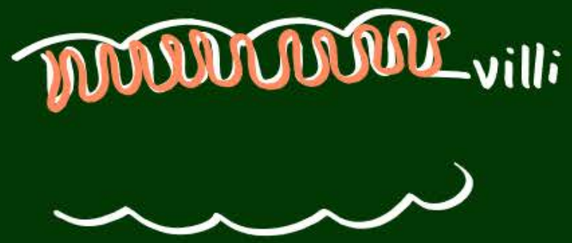
smaller

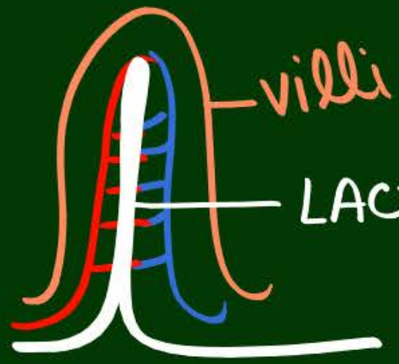
Functions of lymph

i. Nutritive → transportation of Glucose & amino acids to cell.

ii. Drainage

iii. Absorption →





iv. Defence

- Lymph nodes tend to localize the infection and prevent it from spreading to the body as a whole. ✓
- Tonsils on the sides of the neck are also lymph glands.

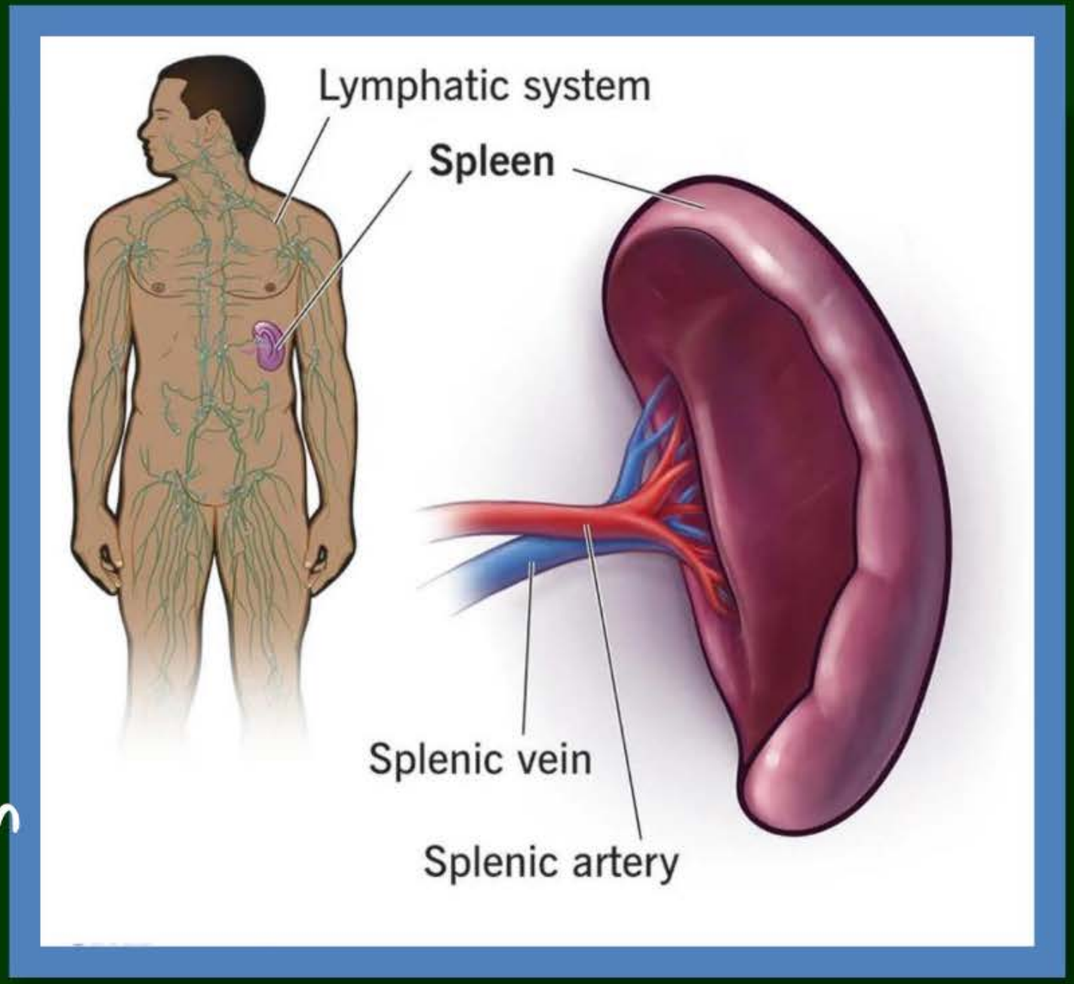
The spleen

- Is a large lymphatic organ
- Size about clenched fist
- Reddish brown in colour and Situated in the abdomen behind the stomach and above the left kidney.



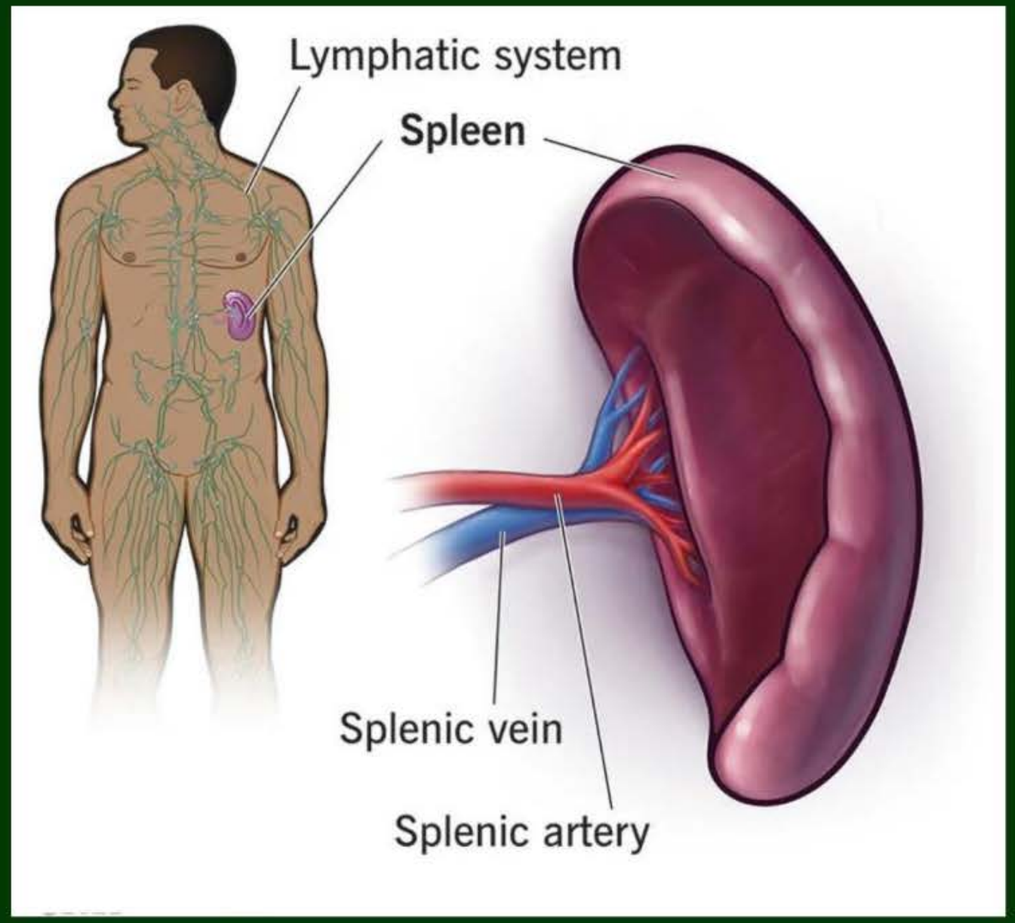
Haem → Bilirubin

Globin



The spleen

- Acts as blood reservoir
- Spleen releases stored blood into blood steam
- Produce lymphocytes
- Destroy worn-out RBCs
- ~~An~~ In embryo, spleen produce RBCs



QUESTION

Name the following:

(a) Any one vein which starts from an organ and ends in another organ besides the heart.

A) Arteriole

B) Venuole

C) Portal Vein ✓

D) Capillaries

QUESTION

TUNICA MEDIA

(b) The kind of blood vessels which have no muscular walls.

A) Arteries

B) Vein

✓ C) Capillaries

D) None of the above

QUESTION

(d) The kind of blood cells which can squeeze out through the walls of one category of blood vessels.

A) RBCs, Capillaries

B) WBCs, Arteries

C) WBC, Capillaries ✓

D) RBC, Veins

QUESTION

(g) The phase of the cardiac cycle in which the auricles contract.

A) Ventricular Systole

B) Auricular Diastole

✓ C) Auricular Systole

D) Joint Diastole

NINJU AS

Next Ch- Excretory Sys.

Blood Grp ___ Half hour

Thank You